

Understanding of Quantum Science and Technology

From Blackbody Radiation to Quantum Information

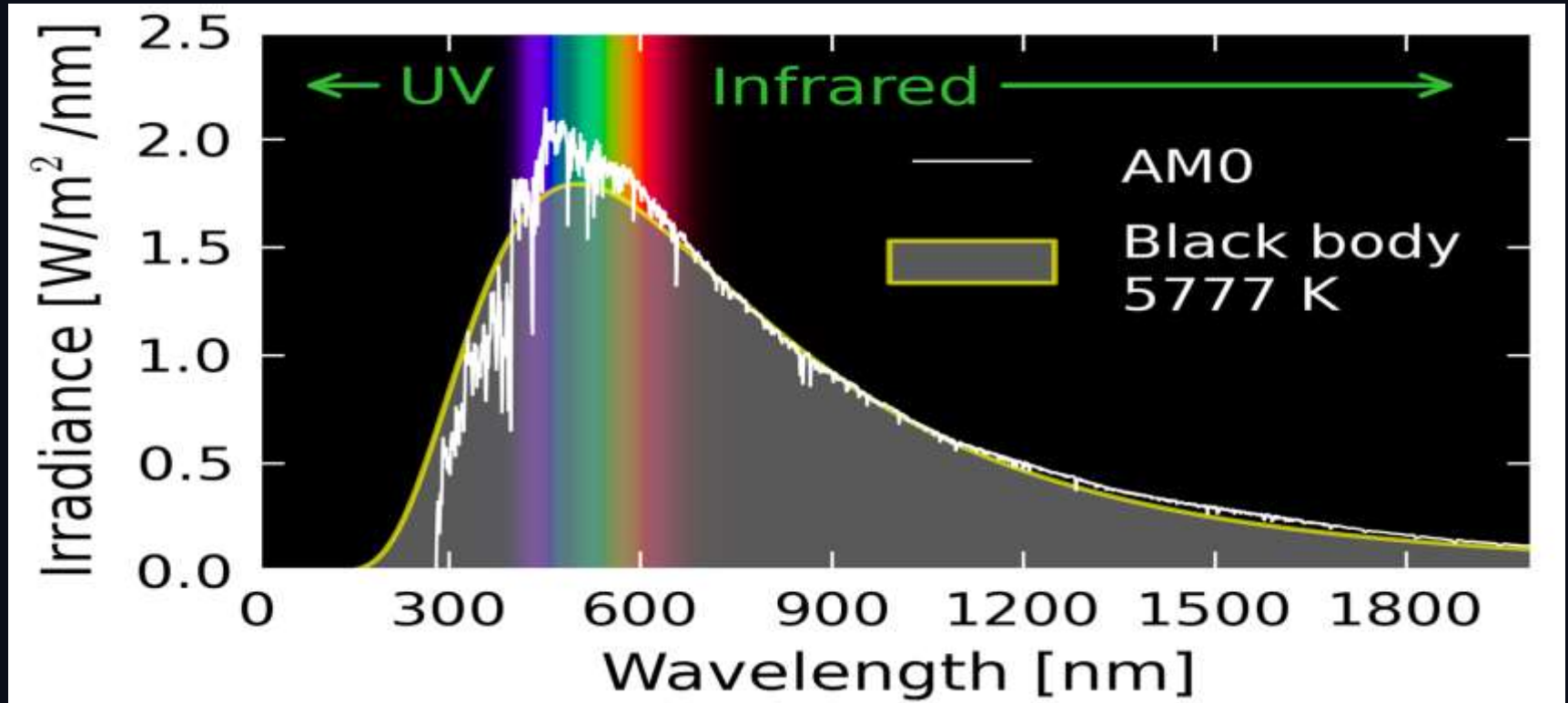
“If you think you understand quantum mechanics, you don’t understand quantum mechanics.”

— *Richard P. Feynman*

Utsav Jignesh Karia
OVSRCS

Origin of Quantum Mechanics:

Light emitted from light bulbs was measured at different temperatures



*Physikalisch-Technische Reichsanstalt (PTR), Berlin
Experimental work: Lummer, Pringsheim, Rubens (1890s)*

Classical Theory of Black-Body Radiation

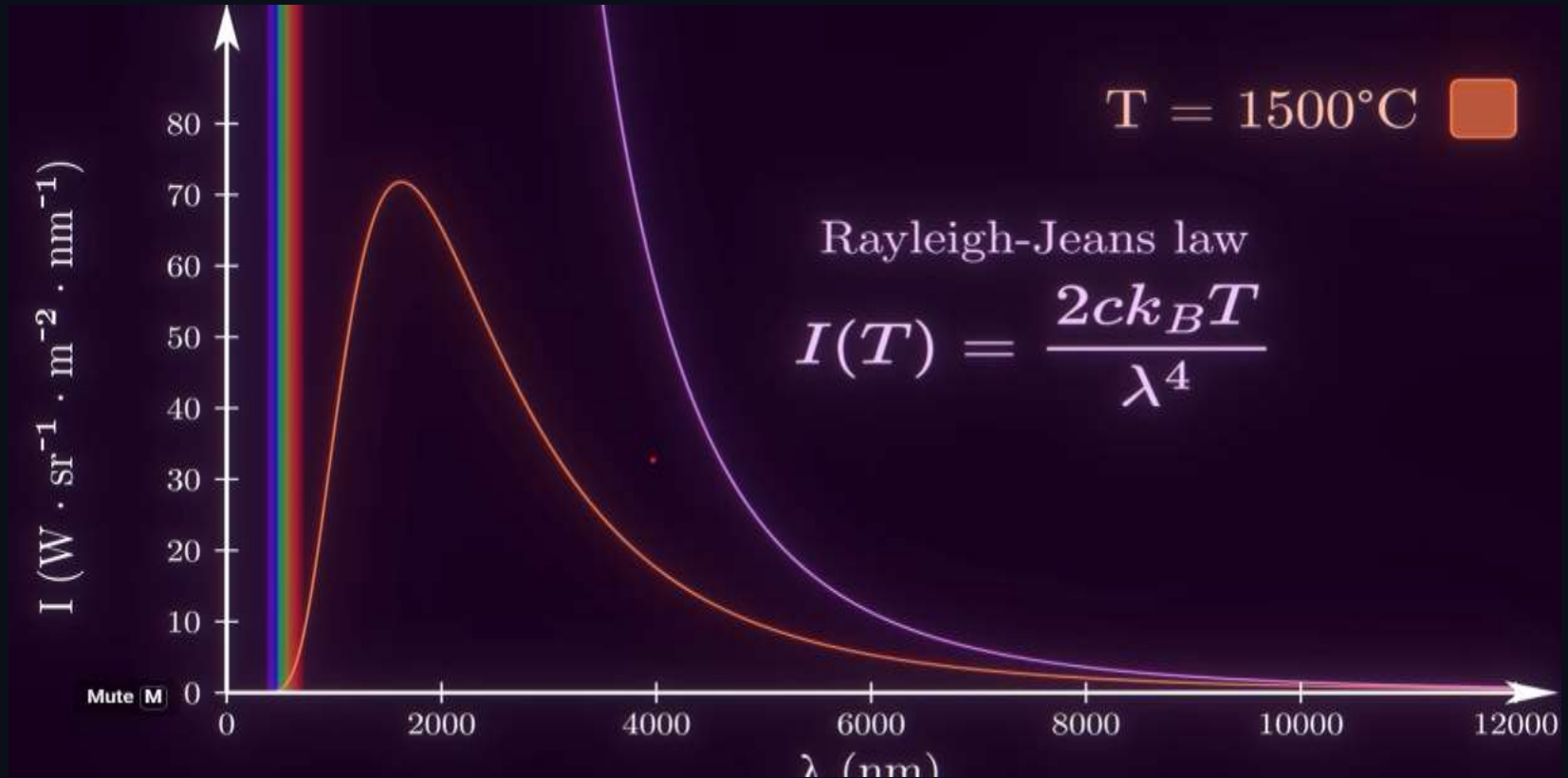
Rayleigh–Jeans Law

$$B_{\lambda}(T) = \frac{2ck_B T}{\lambda^4}$$

- Derived using classical electromagnetism
- Assumes energy is **continuous**
- Matches experiments only at **long wavelengths only**

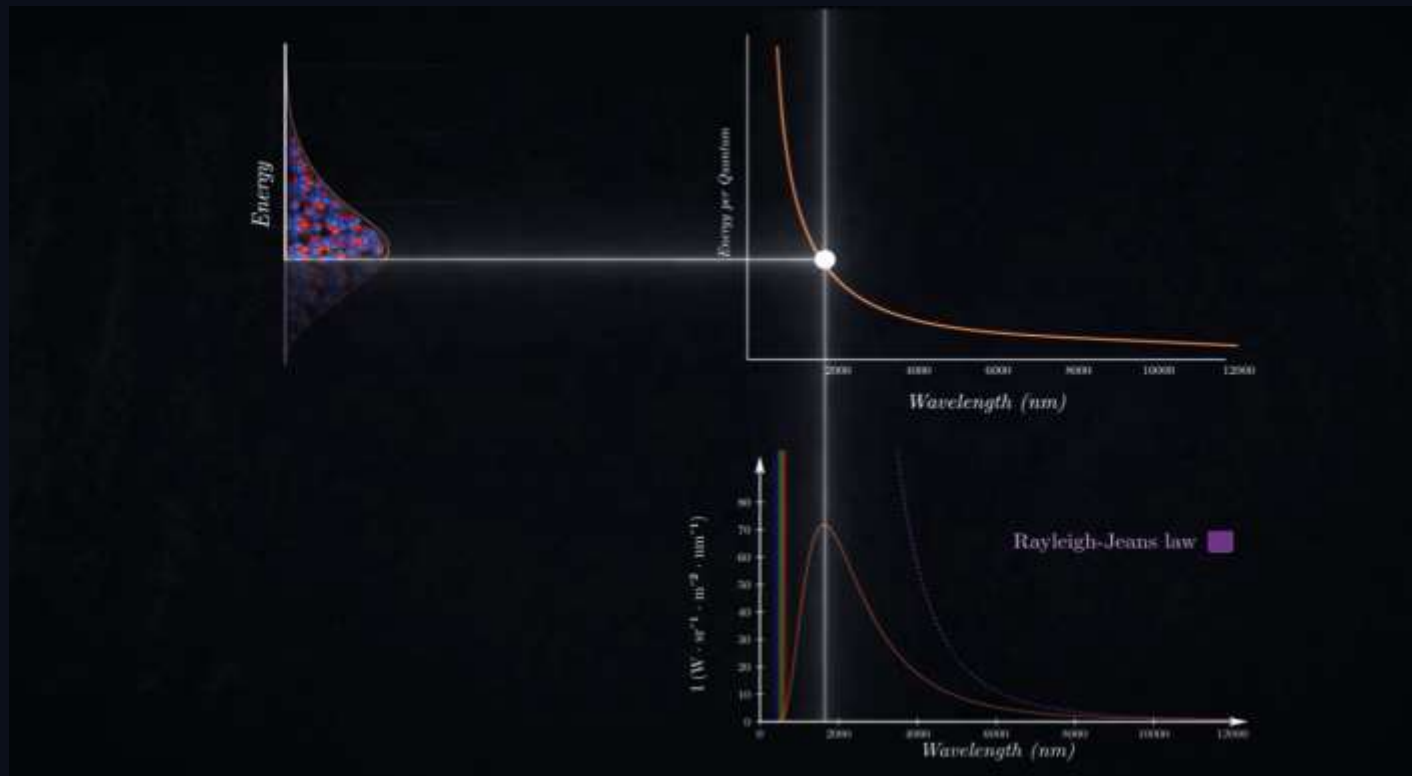
Lord Rayleigh (1900),
Sir James Jeans (1905)

Ultraviolet Catastrophe



Planck's Quantum Hypothesis (1900)

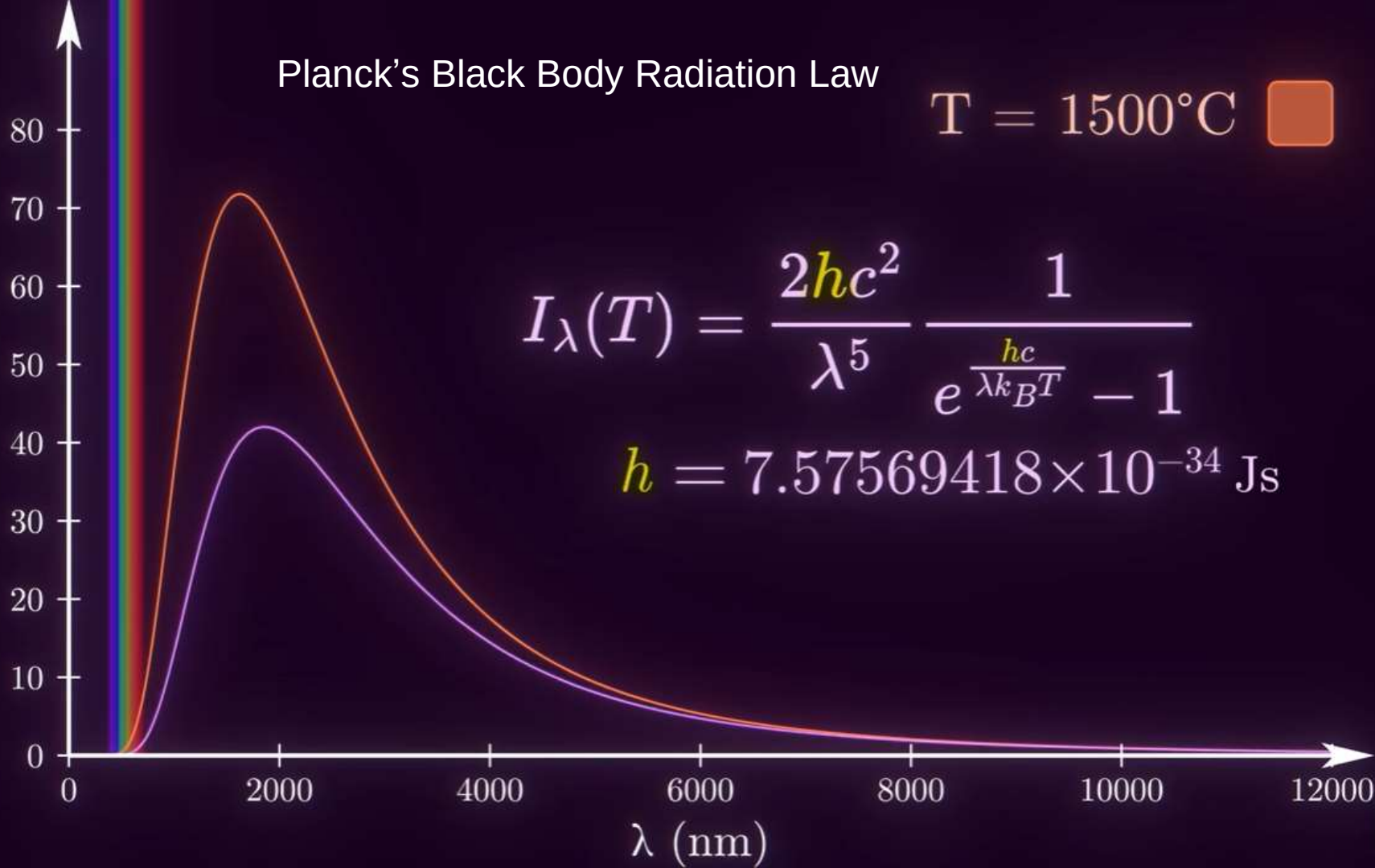
$$E = h\nu$$



Planck's Black Body Radiation Law

$T = 1500^{\circ}\text{C}$ 

$I (\text{W} \cdot \text{sr}^{-1} \cdot \text{m}^{-2} \cdot \text{nm}^{-1})$



From Classical Particles to Quantum Waves

Not only light — **matter itself behaves like a wave**

$$\lambda = \frac{h}{p}.$$

de Broglie wavelength

- Particles described by **wavefunctions**
- Momentum $\leftrightarrow$ wavelength
- Sets the stage for quantum dynamics

Quantum States, Dynamics, and Measurement

State Evolution

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{r}, t) = \hat{H} \psi(\mathbf{r}, t)$$

Schrödinger Equation

Measurement & Probability

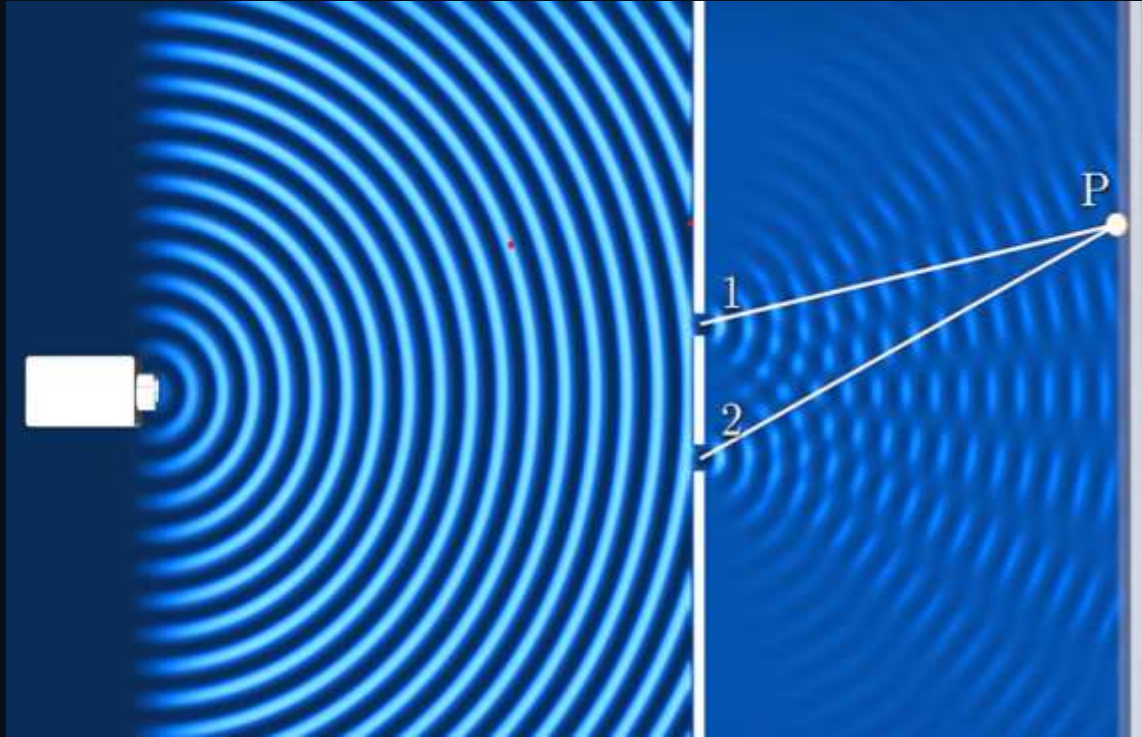
$$P(a) = |\langle a | \psi \rangle|^2$$

Born Rule

- Quantum systems described by **state vectors / wavefunctions**
- Time evolution is **deterministic and unitary**
- Measurement outcomes are **intrinsically probabilistic**

Erwin Schrödinger (1926) ·
Max Born (1926)

Double Slit Experiment



$$\psi = \psi_1 + \psi_2$$

$$P = |\psi_1 + \psi_2|^2$$

Light Takes Every Possible Path

$$\mathcal{A}(A \rightarrow B) = \sum_{\text{all paths}} e^{\frac{i}{\hbar} S[\text{path}]}$$

Feynman Path Integral (1948)

$$S = \int_{t_A}^{t_B} L(x, \dot{x}) dt$$

Action functional

Derivation of Feynman Path Integral

$$A = e^{i(\sum \frac{2\pi}{\lambda} \Delta x - 2\pi f \Delta t)}$$

$$A = e^{i(\sum \frac{2\pi mv}{h} \Delta x - \frac{2\pi E}{h} \Delta t)}$$

$$A = e^{i \frac{\sum mv \Delta x - E \Delta t}{\hbar}}$$

$$A = e^{i \frac{\sum (mv \frac{\Delta x}{\Delta t} - E) \Delta t}{\hbar}}$$

$$E = T + V = \frac{1}{2}mv^2 + V$$

$$A = e^{i \frac{\int \left(\overbrace{mv \frac{dx}{dt}}^{\frac{mv^2}{2}} - \underbrace{E}_{V} \right) dt}{\hbar}}$$

$$A = e^{i \frac{\int \left(mv^2 - \frac{1}{2}mv^2 - V \right) dt}{\hbar}}$$

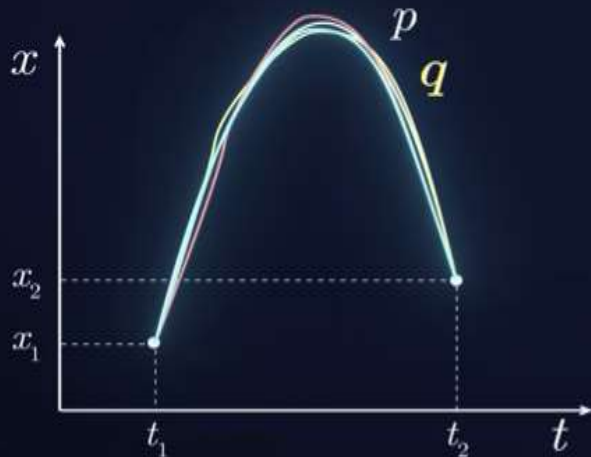
$$A = e^{i \frac{\overbrace{\int \left(\frac{1}{2}mv^2 - V \right) dt}^S}{\hbar}}$$

Light goes through all paths, but we only see Path of Least Action

$$A = e^{i\frac{S}{\hbar}} \quad \hbar \approx 10^{-34} \text{ J}\cdot\text{s}$$

$$dS = S[q] - S[p] = 0$$

$$\frac{dS}{\hbar} \approx 0$$



p : Path of Least Action



Mathematical Structure of Quantum Mechanics

Hilbert Space

$\mathcal{H}$ (complex Hilbert space)

State vectors

$$|\psi\rangle \in \mathcal{H}$$

Operators & observables

$$\hat{A} = \hat{A}^\dagger$$

Every physical observable is represented by a linear Hermitian operator acting on the Hilbert space

Observable

Operator

Position x

$\hat{x} = x$

Momentum p

$\hat{p} = -i\hbar \frac{d}{dx}$

Energy

$\hat{H}$ (Hamiltonian)

Spin

Pauli matrices

Expectation values

$$\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle$$

The **average result** of many measurements on identically prepared systems.

Quantum Computing

Bits

$$b \in \{0, 1\}$$

Properties:

- Definite state
- Deterministic
- No phase
- One value at a time

Qubits

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\alpha|^2 + |\beta|^2 = 1$$

Properties:

- Superposition
- Probabilistic outcomes
- Phase matters
- Interference possible

Quantum Gates and Unitary Evolution

$$|\psi(t)\rangle = U |\psi(0)\rangle$$

Unitary Time Evolution

$$U^\dagger U = I$$

Unitarity preserves probability

Examples

- **Hadamard gate**: creates superposition
- **Pauli-X**: quantum NOT
- **CNOT**: generates entanglement

Quantum Information Theory ·
Dirac · von Neumann

Limits of Quantum Search (BBBV's Theorem)

No quantum algorithm can search an unstructured database in fewer than $\Omega(N)$ queries.

$$Q_{\text{quantum}} \geq c\sqrt{N}$$

- Applies to **unstructured search**
- Rules out exponential speedup
- Grover is **provably optimal**

Bennett · Bernstein · Brassard · Vazirani — 1997

Grover's Algorithm: Geometric Interpretation

Initial Superposition $\Rightarrow$

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle$$

Grover Operator $\Rightarrow$

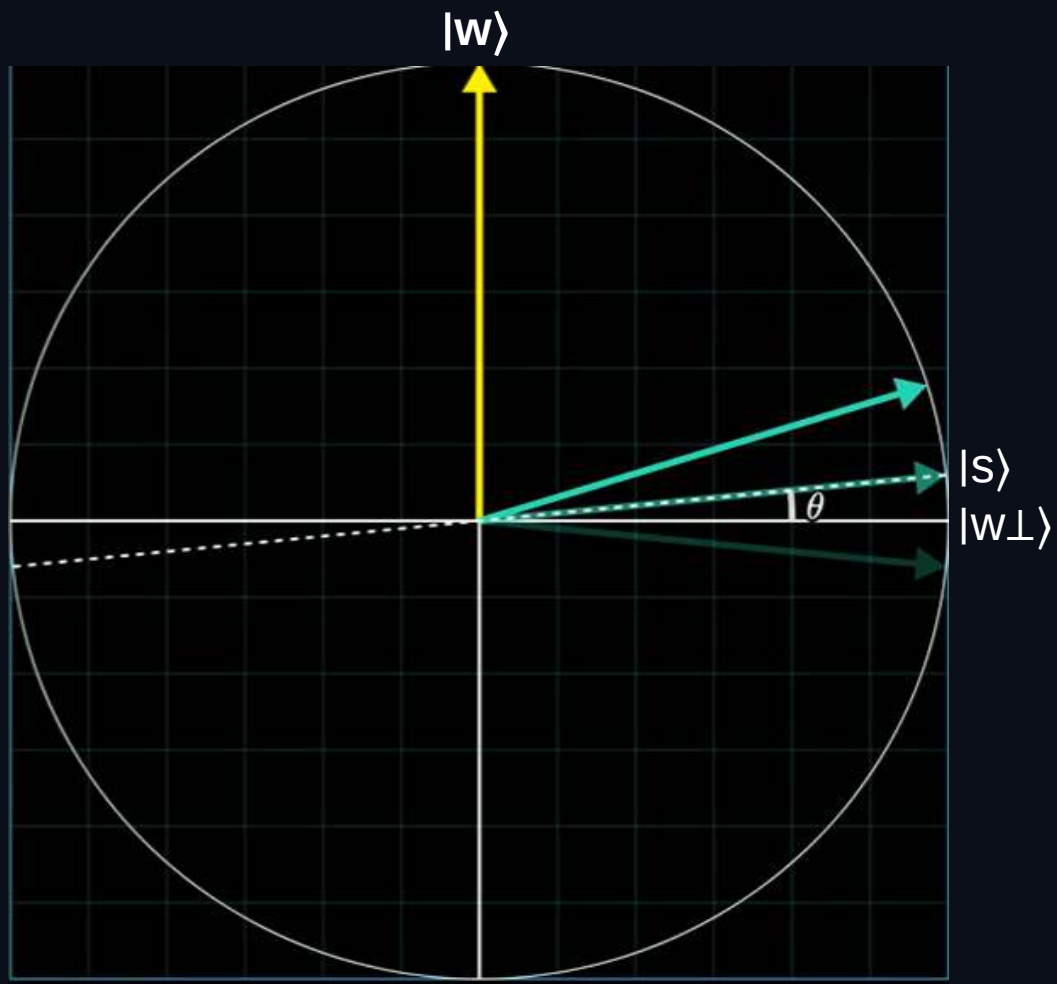
$$G = (2|s\rangle\langle s| - I) O$$

Amplitude evolution $\Rightarrow$

$$|\psi_k\rangle = \sin((2k+1)\theta) |w\rangle + \cos((2k+1)\theta) |w^\perp\rangle$$

Iterations $\Rightarrow$

$$k \approx \frac{\pi}{4} \sqrt{N}$$



O rotates is about $|w\perp\rangle$

$(2|s\rangle\langle s| - I)$ rotates it about $|s\rangle$

Quantum Entanglement

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Bell State

$$|\psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$$

Classical Analogy

- Joint state **cannot be factorized**
- Measurement outcomes are **correlated**
- Resource for quantum technologies

Einstein–Podolsky–Rosen (1935)

Bell (1964)

Non-Separability of Quantum States

$$|\psi_{AB}\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle$$

Entangled (Non-Separable) State

$$|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$$

Separable (Classical-like) State

- Subsystems have **no independent states**
- Only the **joint system** is well-defined
- Correlations exceed classical limits

Schrödinger (1935) · EPR (1935)

Bell Inequalities: Testing Classical Reality

No local hidden-variable theory can reproduce all quantum predictions.

$$|E(a, b) + E(a, b') + E(a', b) - E(a', b')| \leq 2$$

CHSH Inequality

- Classical (local realism): ≤ 2
- Quantum mechanics: **up to $2\sqrt{2}$**
- Experiments violate the classical bound

John S. Bell — 1964

Aspect (1982) · Loophole-free tests (2015)

Entanglement as a Physical Resource

Entanglement enables tasks impossible with classical correlations.

Key applications

- **Quantum teleportation** → transmit an unknown quantum state
- **Superdense coding** → send 2 classical bits using 1 qubit
- **Quantum cryptography (QKD)** → provably secure communication
- **Quantum computing** → speedups via correlated amplitudes

Bennett · Brassard · Popescu · Schumacher

Quantum Teleportation

$$|\psi\rangle \xrightarrow{\text{entanglement} + \text{classical bits}} |\psi\rangle$$

State transfer, not particle transfer

Original state is **destroyed**

Protocol

- Shared **entangled Bell pair**
- **Bell-state measurement** on sender's side
- **Two classical bits** sent
- **Unitary correction** at receiver

Bennett et al. — 1993

Decoherence: The Enemy of Quantum States

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \mathcal{L}(\rho)$$

Loss of coherence

$$\rho \rightarrow \text{off-diagonal terms} \rightarrow 0$$

- Caused by **interaction with environment**
- Destroys **superposition & entanglement**
- Explains quantum $\rightarrow$ classical transition

Quantum Noise Channels

General Quantum Channel

$$\rho' = \mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger$$

Kraus Representation

$$\sum_k E_k^\dagger E_k = I$$

Trace Preservation

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}$$

Amplitude Damping Channel

$$E_0 = \sqrt{1-\lambda}I, \quad E_1 = \sqrt{\lambda}Z$$

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \longrightarrow \begin{pmatrix} \rho_{00} & e^{-\Gamma t} \rho_{01} \\ e^{-\Gamma t} \rho_{10} & \rho_{11} \end{pmatrix}$$

Phase Damping (Dephasing)

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$$

Depolarizing Channel

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right)$$

Lindblad Master Equation

Markovian Open Quantum Dynamics

- Noise modeled as **CPTP maps**
- Decoherence suppresses off-diagonal terms
- Fundamental limit to scalability

Kraus · Lindblad · Gorini–Kossakowski–Sudarshan

Quantum Error Correction

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \longrightarrow \alpha|000\rangle + \beta|111\rangle$$

Logical encoding

- Protects against **decoherence & noise**
- Uses **entanglement + redundancy**
- Requires **fault-tolerant operations**

Shor (1995) · Steane (1996)

Two-Qubit Systems: The Minimal Quantum World

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

General two-qubit pure state

$$\rho = |\psi\rangle\langle\psi|$$

Density Matrix for a
pure Quantum State

$$\rho \longrightarrow \mathcal{E}(\rho)$$

Noisy Version

- Smallest system that allows **entanglement**
- Basis of **teleportation & error correction**
- Decoherence effects become **visible**

Quantum Biology: Quantum Effects in Living Systems

Quantum coherence and entanglement play functional roles in biological processes.

Examples:

- **Photosynthesis** → coherent exciton transport
- **Avian magnetoreception** → radical pair mechanism
- **Enzyme reactions** → quantum tunneling
- **Olfaction (hypothesis)** → vibrational sensing

$$\rho(t) = e^{-iHt/\hbar} \rho(0) e^{iHt/\hbar}$$

Photosynthetic coherence
(density matrix form)

$$\dot{\rho} = -\frac{i}{\hbar} [H_{\text{spin}}, \rho] + \mathcal{L}(\rho)$$

Radical pair spin dynamics

Limits, Open Questions, and the Road Ahead

- How long can **coherence** survive in real systems?
- Can **fault-tolerant quantum computing** scale?
- What are the **ultimate limits** of quantum speedup?
- Does **nature actively protect coherence** (quantum biology)?
- Where does **quantum** → **classical** truly emerge?

Quantum Information · Open Quantum Systems · Complexity Theory

The Big Picture

Quantum mechanics is not just strange — it is fundamental, powerful, and limited.

- Quantum theory explains nature beyond classical intuition
- Interference & entanglement enable new technologies
- Decoherence sets fundamental limits

From light bulbs to quantum computers — the same theory governs it all.

Representing

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Thank you
Questions welcome